GENERALIZED HERMITE–BIRKHOFF INTERPOLATION OF ARBITRARY ORDER IN ANY DIMENSION VIA SINGULAR-WEIGHT MOVING LEAST SQUARES

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Abstract. [**TODO:** write last.] Singular-weight interpolating moving least squares (SWIMLS) reproduces function values at scattered nodes. We study the method when each node carries an arbitrary finite set of linear differential data $q(D)f(x_i)$. We prove that a datum is reproduced exactly if (i) every derivative of its symbol q lies in the span of the symbols prescribed at that node and (ii) $\Delta^{\alpha/2}q = 0$, where α is the (even) order of the weight singularity, and that (ii) cannot be dropped. For classical Hermite data of order k this shows that $\alpha > k$ suffices, sharpening a conjecture of Komargodski and Levin; we also give an exact counterexample to their claim that derivative data at points without function values are interpolated, and a completion procedure that makes arbitrary Birkhoff data exact.

1. Introduction

[**TODO:** Context: MLS (McLain, Lancaster–Šalkauskas), interpolating MLS with singular weights, Shepard; Hermite data in meshless methods; multivariate Hermite–Birkhoff interpolation and poisedness (Gasca–Sauer); Komargodski–Levin. Contributions as a list. Outline.] Contributions.

(1) A structure theorem for SWIMLS near a node (Theorem 4.1) and an exact representation of the error in any differential datum (Theorem 4.3).
(2) A sufficient condition for exact reproduction of a datum $q(D)f(x_i)$: closure of the symbol under differentiation and $\Delta^{\alpha/2}q = 0$ (Corollary 4.4); the second condition is necessary (Proposition 4.6).
(3) For classical Hermite data of order k: exactness iff $\alpha > k$ for the full data set, and an explicit description of which individual derivatives are reproduced when $\alpha \leq k$ (Corollary 4.5); this proves a sharpened form of the conjecture of [1].
(4) An exact counterexample to the extension claimed in [1, §3] (Proposition 5.1), and a completion procedure that makes arbitrary Birkhoff data exact (Proposition 5.2).
(5) A numerically stable evaluation of SWIMLS, including the closed-form limit at the nodes, and numerical experiments in one, two and three dimensions.

2. A one-dimensional example

[**TODO:** About one page: nodes $0, 1, 2, 3$, values and slopes, $\alpha = 2$, local quadratics; shape functions φ_0, ψ_0 with the Hermite cardinal property; second derivatives need $\alpha = 4$; slopes at points without values are not reproduced and completion repairs them. Figure 1.]

3. Setting and notation

3.1. **Polynomials, symbols and the Fischer pairing.** For $d \geq 1$ and $m \geq 0$ let $\Pi_m = \Pi_m(\mathbb{R}^d)$ be the real polynomials of total degree at most m. Multi-indices are $\beta \in \mathbb{N}_0^d$, with

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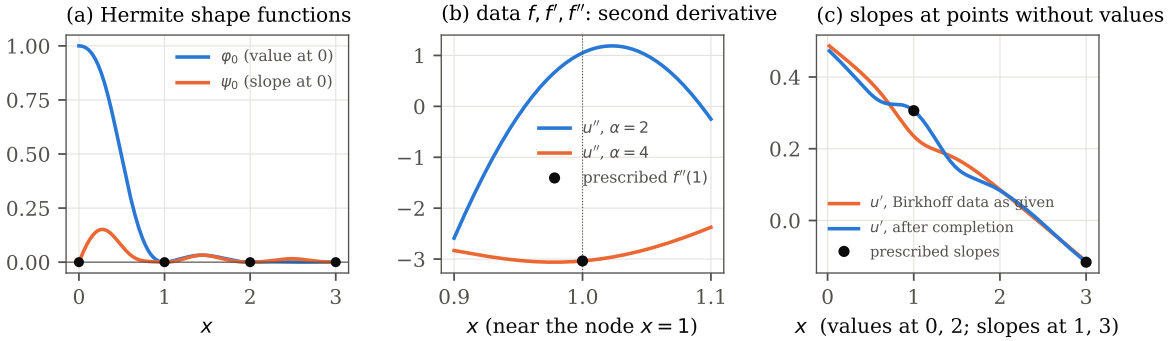


FIGURE 1. The one-dimensional example. (a) Shape functions for the value and the slope at $x = 0$ (nodes $0, 1, 2, 3$, $m = 2$, $\alpha = 2$). (b) Data f, f', f'' at the nodes: u'' near the node $x = 1$ for $\alpha = 2$ and $\alpha = 4$; only $\alpha = 4$ reproduces $f''(1)$ (Corollary 4.5). (c) Values at $0, 2$ and slopes at $1, 3$: the slopes are not reproduced (Proposition 5.1) until the data are completed (Proposition 5.2).

$|\beta| = \sum_j \beta_j$, $\beta! = \prod_j \beta_j!$, $D^\beta = \partial^{|\beta|}/\partial x_1^{\beta_1} \cdots \partial x_d^{\beta_d}$. A polynomial $q(\xi) = \sum_\beta q_\beta \xi^\beta$, called a *symbol*, defines the differential operator $q(D) = \sum_\beta q_\beta D^\beta$. We write $\partial^\gamma q$ for derivatives of a symbol with respect to ξ, and $\Delta q = \sum_j \partial_j^2 q$.

For a symbol q, a smooth function g and a point z we set

$$[q, g]_z := \big(q(D)g\big)(z), \qquad [q, g] := [q, g]_0. \tag{1}$$

For polynomials q, p, $[q, p] = \sum_\beta \beta!\, q_\beta p_\beta$ is the *Fischer inner product*; it is positive definite on Π_m. Two elementary facts will be used repeatedly.

Lemma 3.1 (Fischer adjointness)**.** *For symbols q and polynomials p, $[q, z_j p] = [\partial_j q, p]$. Consequently $[q, |z|^{2\ell} p] = [\Delta^\ell q, p]$ for every $\ell \in \mathbb{N}_0$, where $|z|^2 = \sum_j z_j^2$.*

Proof. With e_j the j-th unit multi-index, $[q, z_j p] = \sum_\kappa \kappa!\, q_\kappa p_{\kappa - e_j} = \sum_\lambda (\lambda + e_j)!\, q_{\lambda + e_j} p_\lambda = \sum_\lambda \lambda!\, (\lambda_j + 1) q_{\lambda + e_j} p_\lambda = [\partial_j q, p]$. Multiplication by $|z|^2 = \sum_j z_j \cdot z_j$ is therefore adjoint to $\sum_j \partial_j^2 = \Delta$, and the second statement follows by induction. □

Lemma 3.2 (Taylor expansion of operators)**.** *Let $A = (A_1, \dots, A_d)$ and $B = (B_1, \dots, B_d)$ be commuting operators. For every symbol q, $q(A + B) = \sum_\gamma \frac{1}{\gamma!} A^\gamma (\partial^\gamma q)(B)$ (a finite sum).*

Proof. This is Taylor's formula $q(a + b) = \sum_\gamma \frac{a^\gamma}{\gamma!} (\partial^\gamma q)(b)$ for the polynomial q, an identity in the commutative polynomial ring in a, b. □

Lemma 3.3. *Let $\ell \in \mathbb{N}_0$ and $\beta \in \mathbb{N}_0^d$. Then $\Delta^\ell \xi^\beta = 0$ if and only if $\sum_j \lfloor \beta_j / 2 \rfloor < \ell$.*

Proof. By the multinomial formula, $\Delta^\ell \xi^\beta = \sum_{|\kappa| = \ell} \frac{\ell!}{\kappa!}\, \partial^{2\kappa} \xi^\beta$, and $\partial^{2\kappa} \xi^\beta = \frac{\beta!}{(\beta - 2\kappa)!} \xi^{\beta - 2\kappa}$ if $2\kappa \le \beta$ and 0 otherwise. Distinct κ give distinct monomials and all coefficients are positive, so $\Delta^\ell \xi^\beta \neq 0$ if and only if some κ with $|\kappa| = \ell$ satisfies $2\kappa \le \beta$, i.e. $\sum_j \lfloor \beta_j / 2 \rfloor \ge \ell$. □

3.2. **Data.** Let $\Omega \subset \mathbb{R}^d$ be open and $X = \{x_1, \dots, x_K\} \subset \Omega$ distinct nodes. Node x_i carries symbols $q_{i,1}, \dots, q_{i,n_i}$ of degree at most m and values $y_{i,1}, \dots, y_{i,n_i} \in \mathbb{R}$, i.e. the interpolation conditions

$$\lambda_{i,s}(f) := [q_{i,s}, f]_{x_i} = \big(q_{i,s}(D) f\big)(x_i) = y_{i,s}, \qquad s = 1, \dots, n_i,\ i = 1, \dots, K. \tag{2}$$

We write $Q_i = \operatorname{span}\{q_{i,1}, \dots, q_{i,n_i}\}$ for the space of symbols at x_i. Every $q = \sum_s c_s q_{i,s} \in Q_i$ has a prescribed value $y_q := \sum_s c_s y_{i,s}$; we say that the datum q is *reproduced* by a function u if $\big[q, u\big]_{x_i} = y_q$. Classical Hermite data of order k correspond to $Q_i = \operatorname{span}\{\xi^\beta : |\beta| \le k\}$; any other choice will be called Hermite–Birkhoff data.

Assumption 3.4. (A1) $\alpha \in 2\mathbb{N}$ is even, and $w_i(x) = \rho_i(x)\,|x - x_i|^{-\alpha}$ with $\rho_i \in C^\infty(\Omega)$, $\rho_i > 0$.
(A2) For every i the symbols $q_{i,1}, \dots, q_{i,n_i}$ are linearly independent.
(A3) (Unisolvency) If $p \in \Pi_m$ and $\lambda_{i,s}(p) = 0$ for all i, s, then $p = 0$.

Since $\big[\cdot,\cdot\big]_{x_i}$ is non-degenerate on Π_m (it is the Fischer pairing after translation) and $\deg q_{i,s} \le m$, (A2) is equivalent to the linear independence of the functionals $\lambda_{i,1}, \dots, \lambda_{i,n_i}$ on Π_m.

Definition 3.5 (SWIMLS)**.** For $x \in \Omega \setminus X$ let

$$J_x(p) = \sum_{i=1}^{K} w_i(x) \sum_{s=1}^{n_i} \big(\lambda_{i,s}(p) - y_{i,s}\big)^2, \qquad p_x := \operatorname*{arg\,min}_{p \in \Pi_m} J_x(p), \tag{3}$$

and define the interpolant $u(x) := p_x(x)$.

By (A1) and (A3), J_x is a strictly convex quadratic on Π_m for every $x \notin X$, so p_x is unique; it depends smoothly on $x \in \Omega \setminus X$ because the coefficients of J_x do and its Hessian is positive definite. Hence $u \in C^\infty(\Omega \setminus X)$. The behaviour at the nodes is the subject of the next section.

Remark 3.6. Positive weights for the individual functionals of a node, or a weight that is singular only for some of them, can be included without change in the arguments below, as long as all data at x_i share the same singular factor $|x - x_i|^{-\alpha}$. [**TODO:** decide whether to state the theory in this generality.]

4. Main results

Throughout this section we fix a node x_i and write

$$N_i = \{p \in \Pi_m : \lambda_{i,s}(p) = 0,\ s = 1, \dots, n_i\} = \{p \in \Pi_m : \big[q, p\big]_{x_i} = 0\ \forall q \in Q_i\}$$

for the polynomials that the data at x_i do not see.

4.1. Structure near a node.

Theorem 4.1 (Structure near a node)**.** *Under Assumption 3.4 there are a neighbourhood $U \subset \Omega$ of x_i with $U \cap X = \{x_i\}$, a polynomial $p_i^* \in \Pi_m$ with $\lambda_{i,s}(p_i^*) = y_{i,s}$ for all s, and C^∞ maps $\nu : U \to N_i$ and $\sigma : U \to \Pi_m$ such that*

$$p_x = p_i^* + \nu(x) + |x - x_i|^\alpha\, \sigma(x) \qquad \text{for all } x \in U \setminus \{x_i\}. \tag{4}$$

In particular u extends to a function in $C^\infty(\Omega)$, with node value $u(x_i) = \big(p_i^* + \nu(x_i)\big)(x_i)$, where

$$\nu(x_i) = \operatorname*{arg\,min}_{n \in N_i} \sum_{j \ne i} w_j(x_i) \sum_{s=1}^{n_j} \big(\lambda_{j,s}(p_i^* + n) - y_{j,s}\big)^2. \tag{5}$$

Proof. Fix an inner product $\langle \cdot, \cdot \rangle$ on Π_m, let $R = N_i^\perp$, and write $\Lambda_i : \Pi_m \to \mathbb{R}^{n_i}$, $p \mapsto (\lambda_{i,s}(p))_s$ and Λ_j likewise. By (A2) Λ_i is surjective with kernel N_i, so $L := \Lambda_i|_R : R \to \mathbb{R}^{n_i}$ is bijective; put $p_i^* := L^{-1} y_i \in R$. Every $p \in \Pi_m$ is uniquely $p = p_i^* + r + n$ with $r \in R$, $n \in N_i$, and then $\Lambda_i p - y_i = Lr$.

Choose a neighbourhood U of x_i whose closure lies in Ω and contains no other node. For $x \in U \setminus \{x_i\}$ divide J_x by $w_i(x) > 0$:

$$\frac{J_x(p)}{w_i(x)} = |Lr|^2 + t(x)\, B_x(p), \qquad t(x) := |x - x_i|^\alpha, \quad B_x(p) := \sum_{j \neq i} \frac{w_j(x)}{\rho_i(x)} \, |\Lambda_j p - y_j|^2.$$

Since α is even, $t(x) = \big(\sum_k (x_k - x_{i,k})^2\big)^{\alpha/2}$ is a polynomial in x; and $x \mapsto B_x$ is a C^∞ family of convex quadratics on U because U contains no node other than x_i. A polynomial $p = p_i^* + r + n$ minimises J_x if and only if the gradient vanishes (the problem is convex), that is

$$2L^*Lr + t(x)\nabla_R B_x(p) = 0, \qquad t(x)\nabla_N B_x(p) = 0,$$

where ∇_R, ∇_N denote the components of the gradient in R and N_i and L^* is the adjoint of L. Since $t(x) > 0$ for $x \neq x_i$ we may substitute $r = t(x)\, s$ and divide by $t(x)$, obtaining $F(x, t(x); s, n) = 0$ for the map $F : U \times \mathbb{R} \times R \times N_i \to R \times N_i$,

$$F(x, t; s, n) := \Big(2L^*Ls + \nabla_R B_x(p_i^* + ts + n),\ \nabla_N B_x(p_i^* + ts + n)\Big),$$

which is C^∞ in all variables, including t near 0.

The point $(x_i, 0)$. Here $F = 0$ reads $\nabla_N B_{x_i}(p_i^* + n) = 0$ and $s = -(2L^*L)^{-1}\nabla_R B_{x_i}(p_i^* + n)$. The restriction of B_{x_i} to the affine space $p_i^* + N_i$ is strictly convex: its Hessian is $2\sum_{j \neq i} \frac{w_j(x_i)}{\rho_i(x_i)} \Lambda_j^* \Lambda_j$ restricted to N_i, and if $n \in N_i$ satisfies $\Lambda_j n = 0$ for all $j \neq i$ then all data functionals vanish on n, so $n = 0$ by (A3). Hence there is exactly one solution (s_0, n_0), and n_0 is the minimiser in (5).

Implicit function theorem. At $(x_i, 0; s_0, n_0)$ the derivative of F with respect to (s, n) is block triangular,

$$\frac{\partial F}{\partial (s, n)} = \begin{pmatrix} 2L^*L & \big(\nabla^2 B_{x_i}\big)_{RN} \\ 0 & \big(\nabla^2 B_{x_i}\big)_{NN} \end{pmatrix},$$

because every derivative with respect to s of a B-term carries the factor $t = 0$. Both diagonal blocks are invertible (L is injective; the second block is the Hessian above). By the implicit function theorem there are a neighbourhood $U' \times (-\tau, \tau)$ of $(x_i, 0)$ and a C^∞ map $(x, t) \mapsto (s(x,t), n(x,t))$ with $F(x, t; s(x,t), n(x,t)) = 0$. Shrinking U to $U' \cap \{t(x) < \tau\}$, for $x \in U \setminus \{x_i\}$ the polynomial $p_i^* + t(x)s(x, t(x)) + n(x, t(x))$ satisfies the gradient equations of the strictly convex function J_x, hence it equals p_x. Setting $\nu(x) := n(x, t(x)) \in N_i$ and $\sigma(x) := s(x, t(x))$ gives (4); both maps are C^∞ because t is a polynomial. Finally $u(x) = p_x(x)$ is, by (4), the restriction to $U \setminus \{x_i\}$ of the C^∞ function $x \mapsto (p_i^* + \nu(x) + t(x)\sigma(x))(x)$, whose value at x_i is $(p_i^* + \nu(x_i))(x_i)$. □

Remark 4.2. Evenness of α is used exactly once: to make $t(x) = |x - x_i|^\alpha$ smooth at x_i. For odd α the same argument gives (4) with ν, σ smooth functions of $(x, |x - x_i|^\alpha)$, and u is in general only finitely smooth at the nodes. If the ρ_i are real analytic, the analytic implicit function theorem shows that u is real analytic.

4.2. **An exact error representation.**

Theorem 4.3 (Error representation)**.** *With the notation of Theorem 4.1, let* $\tau(z) := \sigma(z)(z)$ *for* $z \in U$ *(a* C^∞ *scalar function). For every symbol* q*,*

$$[q, u]_{x_i} = [q, p_i^*]_{x_i} + \sum_\gamma \frac{1}{\gamma!} D^\gamma \Big[x \mapsto [\partial^\gamma q, \nu(x)]_{x_i}\Big](x_i) + [\Delta^{\alpha/2} q, \tau]_{x_i}. \tag{6}$$

Proof. On $U \times \mathbb{R}^d$ consider the C^∞ function $\Phi(x,y) = p_i^*(y) + \nu(x)(y) + t(x)\,\sigma(x)(y) =: \Phi_1+\Phi_2+\Phi_3$, which is a polynomial in y and satisfies $u(z) = \Phi(z,z)$ for $z \in U$ by Theorem 4.1. By the chain rule along the diagonal, $D_z^\beta[\Phi(z,z)] = \big[(D_x + D_y)^\beta \Phi\big](z,z)$, hence $\big[q,u\big]_{x_i} = \big[q(D_x+D_y)\Phi\big](x_i,x_i)$. We treat the three parts.

Φ_1 does not depend on x, so it contributes $(q(D)p_i^*)(x_i) = \big[q,p_i^*\big]_{x_i}$.

For Φ_2, Lemma 3.2 with $A = D_x$, $B = D_y$ gives $q(D_x+D_y)\Phi_2 = \sum_\gamma \frac{1}{\gamma!} D_x^\gamma\,(\partial^\gamma q)(D_y)\Phi_2$. Evaluating at $y = x_i$ commutes with D_x, and $\big((\partial^\gamma q)(D_y)\Phi_2\big)(x,x_i) = \big[\partial^\gamma q, \nu(x)\big]_{x_i}$; evaluating at $x = x_i$ gives the middle term of (6).

For Φ_3 we use the diagonal again: its contribution is $\big[q, t\,\tau\big]_{x_i} = \big(q(D)[\,|z-x_i|^\alpha \tau(z)]\big)(x_i)$. Translate x_i to 0 and let $T\tau$ be the Taylor polynomial of τ at 0 of degree $\deg q$. Since $|z|^\alpha(\tau - T\tau)$ vanishes at 0 to order larger than $\deg q$, it is annihilated by $q(D)$ at 0, and by Lemma 3.1

$$\big[q, |z|^\alpha \tau\big] = \big[q, |z|^\alpha T\tau\big] = \big[\Delta^{\alpha/2} q, T\tau\big] = \big[\Delta^{\alpha/2} q, \tau\big],$$

the last equality because $\deg \Delta^{\alpha/2} q \le \deg q$. □

4.3. **Exact reproduction.**

Corollary 4.4 (Sufficient condition)**.** *Let Assumption 3.4 hold and let $q \in Q_i$. If*

(i) $\partial^\gamma q \in Q_i$ for every multi-index γ, and

(ii) $\Delta^{\alpha/2} q = 0$,

then the datum q is reproduced: $\big[q,u\big]_{x_i} = y_q$.

Proof. In (6), $\big[q,p_i^*\big]_{x_i} = \sum_s c_s \lambda_{i,s}(p_i^*) = y_q$. By (i), every $\partial^\gamma q$ (including $\gamma = 0$) lies in Q_i, and $\nu(x) \in N_i$, so $\big[\partial^\gamma q, \nu(x)\big]_{x_i} = 0$ for all $x \in U$: the middle term vanishes. The last term vanishes by (ii). □

Condition (i) concerns the datum and the data at its own node only; it holds for every datum when Q_i is *D-invariant* (closed under differentiation), the classical setting of multivariate Hermite interpolation [**TODO:** cite de Boor–Ron, Sauer]. Condition (ii) holds whenever $\deg q < \alpha$, but also, for instance, for every harmonic symbol when $\alpha = 2$.

Corollary 4.5 (Hermite data)**.** *Let x_i carry classical Hermite data of order $k \le m$, i.e. $Q_i = \operatorname{span}\{\xi^\beta : |\beta| \le k\}$. Then the derivative datum $D^\beta f(x_i)$ is reproduced whenever $\sum_j \lfloor \beta_j/2 \rfloor < \alpha/2$. In particular*

(a) all data at x_i are reproduced if $\alpha > k$; the smallest admissible weight exponent is the smallest even integer exceeding k;

(b) for $\alpha = 2$, exactly the multilinear derivatives ($\beta_j \le 1$ for all j, e.g. $\partial_x\partial_y$, $\partial_x\partial_y\partial_z$) are covered, whatever their order.

Proof. Q_i is D-invariant, so (i) holds, and (ii) is Lemma 3.3 with $\ell = \alpha/2$. If $|\beta| \le k < \alpha$ then $\sum_j \lfloor \beta_j/2 \rfloor \le |\beta|/2 < \alpha/2$. □

Corollary 4.5(a) proves, in a sharper form, the conjecture of [1, §3] that $\alpha = 2k$ suffices for data up to order k: already $\alpha = k+1$ (or $k+2$, to make it even) suffices. The next result shows that it cannot be improved: condition (ii) is necessary.

Proposition 4.6 (Necessity of (ii), leading order)**.** *In addition to Assumption 3.4, assume that the data at the nodes other than x_i are unisolvent on Π_m by themselves. Let $q \in Q_i$ satisfy (i), with $\deg q = \alpha$ and $\Delta^{\alpha/2} q \neq 0$. Then for every choice of the data at x_i there are data at the other nodes for which $\big[q,u\big]_{x_i} \neq y_q$.*

Proof. By (i) the middle term of (6) vanishes, and $\Delta^{\alpha/2}q = c$ is a non-zero constant, so the error is $c\,\tau(x_i) = c\,\sigma(x_i)(x_i) = c\,s_0(x_i)$ with s_0 from the proof of Theorem 4.1. We show that $s_0(x_i) \neq 0$ for suitable data.

Since $\deg q = \alpha \geq 2$, some derivative $\partial^\gamma q$ with $|\gamma| = \alpha$ is a non-zero constant, so $1 \in Q_i$ by (i): point evaluation $\ell(p) = p(x_i)$ is a datum at x_i. It vanishes on N_i and is non-zero on Π_m, hence non-zero on R.

Keep y_i fixed and choose $y_j = \Lambda_j(p_i^* + v)$, $j \neq i$, for some $v \in \Pi_m$. Then $B_{x_i}(p) = \langle K(p - p_i^* - v), p - p_i^* - v\rangle$ with $K = \sum_{j\neq i} \frac{w_j(x_i)}{\rho_i(x_i)} \Lambda_j^* \Lambda_j$, which is positive definite on Π_m by the additional assumption. The minimiser n_0 over N_i is characterised by $K(n_0 - v) \perp N_i$, i.e. $K(n_0 - v) \in R$, so $\nabla_R B_{x_i}(p_i^* + n_0) = 2K(n_0 - v)$ and $s_0 = (L^*L)^{-1}Kw$ with $w = v - n_0$. As v ranges over Π_m, w ranges over the K-orthogonal complement $W = \{w : Kw \in R\}$ of N_i, so Kw ranges over all of R and s_0 over all of R. Choosing v with $\ell(s_0) \neq 0$ completes the proof. □

Conjecture 4.7 (Characterisation)**.** *Under Assumption 3.4, and for data in general position, a datum $q \in Q_i$ is reproduced if and only if (i) and (ii) hold.*

Remark 4.8. The conjecture is supported by the experiments of Section 7: the predicted verdict agreed with the computed one for all 175 data tested (classical Hermite nodes of orders up to 7 in one, up to 4 in two and up to 3 in three dimensions with $\alpha \in \{2, 4, 6\}$, a node carrying the harmonic symbols of degree ≤ 3, and Birkhoff patterns). The missing parts of a proof are necessity of (ii) when $\deg q > \alpha$, which requires control of derivatives of τ at x_i, and necessity of (i), which involves the derivatives of ν and hence of the weights of the other nodes. [**TODO:** attempt both; at least necessity of (i) for first-order failures.]

5. Derivative data without function values

5.1. **A counterexample.** Komargodski and Levin [1, §3] state that their Hermite MLS, with weights $|x - z|^{-\alpha}$, also interpolates directional derivatives prescribed at points z_j' that differ from the points carrying function values, the proof being "straightforward … by slight modifications" of the case of co-located data. Condition (i) fails for such data: the symbol of a first derivative differentiates to the constant 1, which is not in Q_i when no value is prescribed at the node. The following exact computation shows that the claim is indeed false.

Proposition 5.1. *Let $d = 1$, $m = 1$ (local polynomials $a + by$, the smallest space allowed in [1]), $\alpha = 2$, $\rho_i \equiv 1$, and prescribe $f(0) = f(2) = 0$ and $f'(1) = f'(3) = 1$. Then*

$$u(x) = \frac{x\,(x-1)(x-2)\,(x^2 - 4x + 5)}{2x^4 - 14x^3 + 37x^2 - 42x + 19},$$

which satisfies $u(0) = u(2) = 0$ but $u'(1) = -1$ and $u'(3) = 31/25$.

Proof. The normal equations of (3) for the two coefficients are a 2×2 linear system with rational coefficients in x; solving it and evaluating $a + bx$ gives the stated u (the denominator has no real zeros). The values and derivatives follow by direct computation. [**TODO:** display the normal equations; script `code/proofs/kl_counterexample.py` reproduces the computation in exact arithmetic.] □

The same failure occurs in two dimensions, with gradients at points without values, for local polynomial degrees 1 and 2 and $\alpha \in \{2, 4\}$ (Section 7).

5.2. **Completion.** Given data at x_i with symbol space Q_i, let $\overline{Q}_i$ be the smallest D-invariant space containing Q_i, and choose a basis $\tilde{q}_{i,1}, \dots, \tilde{q}_{i,r}$ of a complement of Q_i in $\overline{Q}_i$.

Proposition 5.2 (Completion)**.** *Augment the data at x_i by the functionals $\tilde{q}_{i,l}(D)f(x_i)$ with arbitrary values $\tilde{y}_{i,l}$. If Assumption 3.4 holds for the original data and $\deg \overline{Q}_i \le m$, it holds for the augmented data, and every original datum $q \in Q_i$ with $\Delta^{\alpha/2}q = 0$ is reproduced by the augmented interpolant, whatever the values $\tilde{y}_{i,l}$.*

Proof. The augmented symbols at x_i form a basis of $\overline{Q}_i$, so (A2) holds; (A3) is preserved by adding data. Every $q \in Q_i \subset \overline{Q}_i$ satisfies (i) with respect to the D-invariant space $\overline{Q}_i$, and Corollary 4.4 applies. □

The values $\tilde{y}_{i,l}$ influence the accuracy, not the exactness. In the experiments they are estimated by a smooth (non-singular) moving least-squares fit of the given data.

6. Numerical evaluation

[**TODO:** centred, scaled basis; log-normalised weights; SVD least squares; the two-block reformulation near a node; closed-form node value (5); comparison with ε-regularised normal equations (Figure: e5_stability).]

7. Numerical experiments

[**TODO:** exactness experiments (verification method: high-precision stencils at two radii); Birkhoff patterns; completion; convergence (state only what the parameter study supports).]

8. Conclusions

[**TODO:** summary; open problems: Conjecture 4.7; convergence orders; better estimators for completion.]

References

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